Use LikertMakeR to replicate data from SEM results in a published report

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Abstract

This paper demonstrates how the **LikertMakeR** package can reconstruct a realistic survey dataset using only published summary statistics. Using statistics reported by Dwivedi et al. (2019) — including means, standard deviations, correlations, and reliability coefficients—we generate synthetic Likert-scale responses for 340 observations and 43 survey items. The reconstructed dataset closely reproduces the reported descriptive statistics and correlation structure, and when analysed with structural equation modelling it yields relationships similar to those reported in the original study. This example illustrates how LikertMakeR can be used to approximate original datasets from published results, enabling replication exercises, teaching demonstrations, and methodological exploration when raw data are unavailable.

TL;DR

- **Goal:** Demonstrate how the **LikertMakeR** package can reconstruct survey data using only published summary statistics.
- **Source study:** *Dwivedi et al. (2019)*, which reports results from a 43-item survey of 340 respondents measuring 15 constructs.
- **Method:**
  - Generate synthetic scale scores with the same means and standard deviations as the published data.
  - Apply the reported correlation matrix to reproduce the relationships among variables.
  - Reconstruct plausible individual survey items using reported reliability (Cronbach's α).
- **Validation:** The synthetic dataset closely reproduces the published descriptive statistics, correlations, and scale reliabilities.
- **Result:** Structural equation modelling of the synthetic data produces relationships broadly consistent with those reported in the original paper.
- **Takeaway:** *LikertMakeR* can approximate realistic datasets from published results, enabling replication exercises, teaching examples, and methodological exploration when raw data are unavailable.

Dwiveli et al. (2019) data replication

Background

Researchers often publish **summary statistics** - means, standard deviations, correlations, and reliability estimates - without releasing the original data.

In teaching, simulation studies, and methodological work, it can be useful to **reconstruct plausible datasets** from those published summaries.

> ❗ Prerequisites
>
> This tutorial assumes a basic familiarity with:
> - Likert-scale survey data
> - correlation matrices
> - structural equation modelling
>
> Required R packages:
> - `library(LikertMakeR)`
> - `library(lavaan)`

The ***LikertMakeR*** package provides tools for doing exactly this: generating synthetic Likert-scale datasets that reproduce the statistical properties of reported data.

In this tutorial we demonstrate the process using a published study by *Dwivedi et al. (2018)*. Using only the statistics reported in their paper, we reconstruct a synthetic dataset and show that it reproduces the key statistical relationships reported in the original analysis.

Dwivedi A, Johnson LW, Wilkie DC, De Araujo-Gil L (2019). Consumer emotional brand attachment with social media brands and social media brand equity. *European Journal of Marketing*, 53(6): 1176–1204, doi: https://doi.org/10.1108/EJM-09-2016-0511

Dwevedi et al. (2019) sought to understand social media platforms from a "brand" perspective through examining the effect of consumers' emotional attachment on social media consumer-based brand equity (CBBE) using an online survey of 340 Australian social media consumers, as shown in Figure 1.

Overview of the Reconstruction Process

Reconstructing a dataset from published statistics involves several steps:

1. Extract summary statistics from the published paper
2. Reconstruct scale-level distributions (means and standard deviations)
3. Reconstruct the correlation structure between scales
4. Generate synthetic Likert-scale item responses consistent with those statistics
5. Verify that the reconstructed dataset reproduces the reported results

We now walk through each step.

Verifying the Reconstruction

Reconstructing data from summary statistics inevitably involves approximation.

To assess whether the reconstruction is credible, we compare several properties of the synthetic dataset with those reported in the original paper:

- scale means and standard deviations
- inter-scale correlations
- reliability estimates
- structural equation model results

If these statistics closely match the published values, the synthetic data can be considered a reasonable reconstruction of the original dataset.

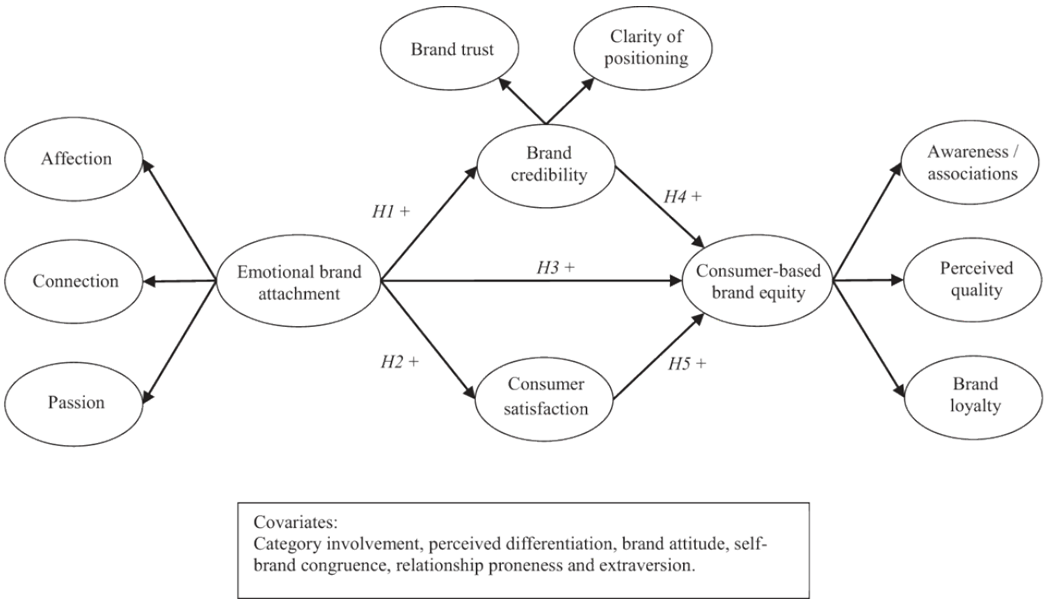

Figure 1: Hypothesised relationships among variables. (*Dwivedi et al. (2019)*)

1 Step 1: Extract summary statistics

To recreate a plausible dataset, we need to match the first two statistical moments of each scale:

- the **mean** (average level of responses), and
- the **standard deviation** (spread of responses).

If synthetic data reproduce these two properties, the reconstructed scales behave similarly to the original ones.

Fifteen constructs were measured with Likert scales constructed from 43 items (questions). Summary statistics from the study are shown in Table 1

2 Step 2: Reconstruct scale distributions

Generate data with same means & standard deviations.

With those summary statistics, we can synthesise a dataframe with the same first and second moments as the original, using the `LikertMakeR::lfast()` function.

```
library(LikertMakeR)

sample_size <- 340
lower <- 1
upper <- 5
```

construct	name	n_items	mean	sd	alpha
Affection (AFF)	AFF	2	2.51	0.93	0.87
Connection (CON)	CON	2	2.79	0.98	0.87
Passion (PAS)	PAS	3	2.62	0.89	0.84
Clarity of Positioning (COP)	COP	3	3.52	0.70	0.80
Brand trust (TRT)	TRT	4	3.30	0.74	0.88
Consumer satisfaction (SAT)	SAT	3	3.87	0.64	0.86
Awareness/associations (AWR)	AWR	5	4.10	0.62	0.87
Quality (QUL)	QUL	4	3.69	0.66	0.84
Loyalty (LOY)	LOY	2	3.56	0.84	0.79
Perceived differentiation (DIF)	DIF	2	3.80	0.78	0.85
Self-brand congruence (FIT)	FIT	2	3.02	0.83	0.82
Extraversion (EXT)	EXT	2	3.09	0.91	0.79
Brand attitude (ATT)	ATT	2	3.50	0.81	0.88
Relationship proneness (REL)	REL	3	2.90	0.83	0.79
Involvement (INV)	INV	4	3.32	0.92	0.79

Table 1: Summary statistics and variable properties. (*Dwivedi et al. (2019)*)

```r
set.seed(42)

base_data <- data.frame(
  AFF = lfast(sample_size, mean = 2.51, sd = 0.93, lower, upper, items = 2),
  CON = lfast(sample_size, mean = 2.79, sd = 0.98, lower, upper, items = 2),
  PAS = lfast(sample_size, mean = 2.62, sd = 0.89, lower, upper, items = 3),
  COP = lfast(sample_size, mean = 3.52, sd = 0.70, lower, upper, items = 3),
  TRT = lfast(sample_size, mean = 3.30, sd = 0.74, lower, upper, items = 4),
  SAT = lfast(sample_size, mean = 3.87, sd = 0.64, lower, upper, items = 3),
  AWR = lfast(sample_size, mean = 4.10, sd = 0.62, lower, upper, items = 5),
  QUL = lfast(sample_size, mean = 3.69, sd = 0.66, lower, upper, items = 4),
  LOY = lfast(sample_size, mean = 3.56, sd = 0.84, lower, upper, items = 2),
  DIF = lfast(sample_size, mean = 3.80, sd = 0.78, lower, upper, items = 2),
  FIT = lfast(sample_size, mean = 3.02, sd = 0.83, lower, upper, items = 2),
  EXT = lfast(sample_size, mean = 3.09, sd = 0.91, lower, upper, items = 2),
  ATT = lfast(sample_size, mean = 3.50, sd = 0.81, lower, upper, items = 2),
  REL = lfast(sample_size, mean = 2.90, sd = 0.83, lower, upper, items = 3),
  INV = lfast(sample_size, mean = 3.32, sd = 0.92, lower, upper, items = 4)
)
```

With this base dataframe, we check the summary statistics and find in Table 2 that the means and standard deviations are the same as in the reported data, to two decimal places.

3 Step 3: Correlations to match published data

These existing synthetic data have random low correlations. We need to correlate the variables according to the reported data. *Dwivedi et al.* reported the correlation presented in Table 3. This is our target correlation matrix.

We apply this correlation matrix to our existing data so that the variables are corre-

	mean	sd
AFF	2.510	0.928
CON	2.790	0.979
PAS	2.621	0.892
COP	3.520	0.700
TRT	3.300	0.738
SAT	3.869	0.640
AWR	4.099	0.620
QUL	3.690	0.661
LOY	3.560	0.841
DIF	3.800	0.781
FIT	3.021	0.830
EXT	3.090	0.910
ATT	3.500	0.812
REL	2.900	0.832
INV	3.321	0.921

Table 2: Synthetic data summary statistics

	AFF	CON	PAS	COP	TRT	SAT	AWR	QUL	LOY	DIF	FIT	EXT	ATT	REL	INV
AFF	1.00	0.79	0.82	0.47	0.50	0.44	0.35	0.44	0.54	0.46	0.58	0.33	0.66	0.62	0.51
CON	0.79	1.00	0.77	0.43	0.49	0.42	0.36	0.41	0.56	0.41	0.61	0.32	0.65	0.61	0.49
PAS	0.82	0.77	1.00	0.46	0.51	0.46	0.40	0.48	0.55	0.45	0.60	0.36	0.73	0.62	0.48
COP	0.47	0.43	0.46	1.00	0.57	0.54	0.39	0.51	0.50	0.37	0.41	0.24	0.49	0.38	0.31
TRT	0.50	0.49	0.51	0.57	1.00	0.61	0.31	0.61	0.55	0.31	0.52	0.23	0.58	0.43	0.33
SAT	0.44	0.42	0.46	0.54	0.61	1.00	0.53	0.67	0.63	0.40	0.47	0.23	0.63	0.43	0.41
AWR	0.35	0.36	0.40	0.39	0.31	0.53	1.00	0.55	0.49	0.47	0.30	0.29	0.46	0.41	0.48
QUL	0.44	0.41	0.48	0.51	0.61	0.67	0.55	1.00	0.49	0.43	0.43	0.27	0.55	0.44	0.34
LOY	0.54	0.56	0.55	0.50	0.55	0.63	0.49	0.49	1.00	0.47	0.55	0.28	0.63	0.53	0.52
DIF	0.46	0.41	0.45	0.37	0.31	0.40	0.47	0.43	0.47	1.00	0.33	0.36	0.46	0.42	0.42
FIT	0.58	0.61	0.60	0.41	0.52	0.47	0.30	0.43	0.55	0.33	1.00	0.27	0.60	0.51	0.42
EXT	0.33	0.32	0.36	0.24	0.23	0.23	0.29	0.27	0.28	0.36	0.27	1.00	0.39	0.47	0.36
ATT	0.66	0.65	0.73	0.49	0.58	0.63	0.46	0.55	0.63	0.46	0.60	0.39	1.00	0.66	0.55
REL	0.62	0.61	0.62	0.38	0.43	0.43	0.41	0.44	0.53	0.42	0.51	0.47	0.66	1.00	0.71
INV	0.51	0.49	0.48	0.31	0.33	0.41	0.48	0.34	0.52	0.42	0.42	0.36	0.55	0.71	1.00

Table 3: Target correlation matrix. (*Dwivedi et al. (2019)*)

	AFF	CON	PAS	COP	TRT	SAT	AWR	QUL	LOY	DIF	FIT	EXT	ATT	REL	INV
AFF	1.00	0.79	0.82	0.47	0.50	0.44	0.35	0.44	0.54	0.46	0.58	0.33	0.66	0.62	0.51
CON	0.79	1.00	0.77	0.43	0.49	0.42	0.36	0.41	0.56	0.41	0.61	0.32	0.65	0.61	0.49
PAS	0.82	0.77	1.00	0.46	0.51	0.46	0.40	0.48	0.55	0.45	0.60	0.36	0.73	0.62	0.48
COP	0.47	0.43	0.46	1.00	0.57	0.54	0.39	0.51	0.50	0.37	0.41	0.24	0.49	0.38	0.31
TRT	0.50	0.49	0.51	0.57	1.00	0.61	0.31	0.61	0.55	0.31	0.52	0.23	0.58	0.43	0.33
SAT	0.44	0.42	0.46	0.54	0.61	1.00	0.53	0.67	0.63	0.40	0.47	0.23	0.63	0.43	0.41
AWR	0.35	0.36	0.40	0.39	0.31	0.53	1.00	0.55	0.49	0.47	0.30	0.29	0.46	0.41	0.48
QUL	0.44	0.41	0.48	0.51	0.61	0.67	0.55	1.00	0.49	0.43	0.43	0.27	0.55	0.44	0.34
LOY	0.54	0.56	0.55	0.50	0.55	0.63	0.49	0.49	1.00	0.47	0.55	0.28	0.63	0.53	0.52
DIF	0.46	0.41	0.45	0.37	0.31	0.40	0.47	0.43	0.47	1.00	0.33	0.36	0.46	0.42	0.42
FIT	0.58	0.61	0.60	0.41	0.52	0.47	0.30	0.43	0.55	0.33	1.00	0.27	0.60	0.51	0.42
EXT	0.33	0.32	0.36	0.24	0.23	0.23	0.29	0.27	0.28	0.36	0.27	1.00	0.39	0.47	0.36
ATT	0.66	0.65	0.73	0.49	0.58	0.63	0.46	0.55	0.63	0.46	0.60	0.39	1.00	0.66	0.55
REL	0.62	0.61	0.62	0.38	0.43	0.43	0.41	0.44	0.53	0.42	0.51	0.47	0.66	1.00	0.71
INV	0.51	0.49	0.48	0.31	0.33	0.41	0.48	0.34	0.52	0.42	0.42	0.36	0.55	0.71	1.00

Table 4: Synthetic data correlation matrix.

lated according to our target correlations, using the `LikertMakeR::lcor()` function.

```
correlated_data <- LikertMakeR::lcor(
  data = base_data,
  target = dwivedi_correlations
)
names(correlated_data) <- varnames
```

And the revised dataframe should be correlated in the same way as the published data. Results are shown in Table 4.

```
synth_correlations <- cor(correlated_data)
dimnames(synth_correlations) <- list(Rows = varnames, Cols = varnames)
```

The synthetic data correlations appear to be the same, rounded to two decimal places, as the target, published correlations.

To quantify how closely the reconstructed correlations match the published ones, we compute the **Frobenius norm** of the difference between the two matrices.

In simple terms, this measures the overall discrepancy between the two correlation tables. A value close to zero indicates that the reconstructed correlations closely match those reported in the paper.

The *Frobenius Norm* is defined as the square root of the sum of the squares of all the matrix entries Equation 1.

```
mat_diff <- dwivedi_correlations - synth_correlations
frob_diff <- matrixcalc::frobenius.norm(mat_diff)
```

$$F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right)^{1/2} \quad (1)$$

Frobenius norm of a matrix.

Calculated *Frobenius Norm* here is 0.0128301, which is very low for a matrix of this size.

4 Step 4: Generate synthetic Likert responses

So far we have reconstructed the statistical properties of the scales. The next step is to generate individual Likert item responses consistent with those statistics.

Given the correlated variables, we also have information on the nature of the items that make each variable in Table 1. This allows us to estimate them too - thus giving us a complete set of responses as we might have received them in the original survey.

We use the `LikertMakeR::makeItemsScale()` function.

```
aff_items <- makeItemsScale(
  scale = correlated_data$AFF,
  lowerbound = 1, upperbound = 5, items = 2,
  alpha = 0.87, summated = FALSE
)
names(aff_items) <- c("aff1", "aff2")
```

Generating the other 14 sets of scale items works the same way.

The first ten observations are presented in Table 5. They are all integer responses to standard 1-5 Likert-scale-type questions.

The resulting dataset contains simulated item responses that follow the same statistical structure as the original study. Most achieved Cronbach's alpha are as desired as we can see in Table 6.

5 Step 5: Reproduce relationship structure analysis

5.1 Structural Equation Model

Now that we have a complete synthetic data set, we can analyse the data to see if we can find the same relationships found by *Dwivedi et al.*

5.2 original results

The original published results from *Dwivedi et al.* are reproduced in Table 7.

5.3 Results from synthetic data

The authors of the original paper used the *AMOS* program, which is a part of ***SPSS***. I haven't used *SPSS/AMOS* for a very long time, so I'll run my analysis using the `lavaan` package for ***R***. By default, *AMOS* uses Maximum Likelihood (ML) estimation, while *lavaan* uses Robust Maximum Likelihood (MLR). With ML specified and bootstrapped confidence interval estimates, we *should* get comparable output.

```
# model definition
model <- "
# first-order constructs
    AFF =~ aff1 + aff2
    CON =~ con1 + con2
```

	aff1	aff2	con1	con2	pas1	pas2	pas3	cop1	cop2	cop3	trt1	trt2	trt3	trt4	sat1
1	2	1	2	3	1	2	2	4	2	3	3	1	2	3	4
2	2	2	2	3	2	3	2	5	3	3	5	3	4	3	5
3	2	2	1	2	1	2	3	5	3	4	4	2	3	3	4
4	5	4	5	4	4	5	5	5	3	4	5	3	5	5	5
5	4	3	4	3	4	5	3	4	2	2	4	2	3	3	4
6	3	3	3	3	3	3	4	4	2	2	4	2	2	3	4
7	2	1	2	2	2	3	3	5	3	4	4	2	4	4	5
8	2	1	4	3	1	1	2	5	3	3	5	3	5	5	5
9	3	3	4	3	3	4	3	5	3	3	4	2	4	3	5
10	4	4	5	4	4	5	3	5	3	5	5	3	5	4	5

	sat2	sat3	awr1	awr2	awr3	awr4	awr5	qul1	qul2	qul3	qul4	loy1	loy2	dif1	dif2
1	2	3	3	1	1	2	3	1	3	3	1	3	2	2	4
2	4	4	5	4	5	5	5	2	4	4	4	4	3	3	5
3	3	4	5	3	5	4	5	2	4	2	4	5	3	3	5
4	4	5	5	4	5	5	5	3	5	4	5	5	5	5	5
5	3	4	5	3	5	4	5	3	5	5	4	5	3	5	5
6	2	3	4	2	4	3	4	2	4	3	4	4	2	1	3
7	3	4	5	3	5	5	5	3	5	5	5	5	3	3	5
8	4	4	5	3	4	5	3	3	5	4	5	5	3	4	5
9	3	4	5	3	4	3	5	3	5	4	3	5	3	2	4
10	4	5	5	3	5	5	5	3	5	5	5	5	5	4	5

	fit1	fit2	ext1	ext2	att1	att2	rel1	rel2	rel3	inv1	inv2	inv3	inv4
1	1	3	3	4	3	3	1	3	3	4	3	2	1
2	2	4	2	3	3	4	2	4	3	5	2	3	4
3	2	4	1	2	3	3	2	1	3	4	1	3	3
4	3	5	2	4	4	5	4	5	5	5	5	4	5
5	3	4	1	3	4	5	2	4	3	4	2	2	4
6	2	3	2	3	4	4	2	4	4	4	1	3	3
7	2	3	2	4	3	3	2	4	2	5	3	2	3
8	2	3	4	5	3	4	2	4	4	4	2	4	2
9	3	4	2	4	3	4	3	5	3	5	3	5	4
10	4	5	3	5	5	5	3	3	5	5	5	3	5

Table 5: First ten rows of our synthetic data - all 43 items

construct	original	synthetic
AFF	0.87	0.870
CON	0.87	0.869
PAS	0.84	0.839
COP	0.80	0.799
TRT	0.88	0.880
SAT	0.86	0.859
AWR	0.87	0.869
QUL	0.84	0.840
LOY	0.79	0.788
DIF	0.85	0.851
FIT	0.82	0.822
EXT	0.79	0.790
ATT	0.88	0.880
REL	0.79	0.789
INV	0.79	0.789

Table 6: Comparison between Cronbach's alphas for original published constructs and synthetic costructs.

Estimated path		95% CI	Hypothesis support
Panel A: Hypothesised model results			
Emotional brand attachment → Brand credibility	0.72	0.60-0.82	Supported
Emotional brand attachment → Consumer satisfaction	0.54	0.42-0.64	Supported
Emotional brand attachment → Consumer-based brand equity (CBBE)	0.21	0.03-0.40	Not supported
Brand credibility → CBBE	0.31	0.11-0.54	Supported
Consumer satisfaction → CBBE	0.63	0.50-0.76	Supported
Total indirect effect			
Emotional brand attachment (EBA) → CBBE	0.57	0.42-0.77	
Specific indirect effects			
EBA → CBBE via Brand credibility	0.09	0.03-0.18	Supported
EBA → CBBE via Consumer satisfaction	0.14	0.09-0.21	Supported
Panel B: Alternative model results			
Emotional brand attachment → Brand credibility	0.74	0.61-0.84	Supported
Emotional brand attachment → Consumer satisfaction	0.56	0.44-0.68	Supported
Emotional brand attachment → Brand loyalty	0.40	0.17-0.61	Supported
Brand credibility → Brand loyalty	0.14	0.07-0.45	Not supported
Consumer satisfaction → Brand loyalty	0.35	0.19-0.53	Supported
Specific indirect effects			
EBA → LOY via Consumer satisfaction	0.12	0.06-0.20	Supported

Table 7: Parameter estimates of the hypothesised and alternative models - Table IV from *Dwivedi et al. (2019)*.

```
    PAS =~ pas1 + pas2 + pas3
    AWR =~ awr1 + awr2 + awr3 + awr4 + awr5
    QUL =~ qul1 + qul2 + qul3 + qul4
    LOY =~ loy1 + loy2
    COP =~ cop1 + cop2 + cop3
    TRT =~ trt1 + trt2 + trt3 + trt4
    SAT =~ sat1 + sat2 + sat3
# second-ordr constructs
    Attachment =~  AFF + CON + PAS
    Credibility =~ TRT + COP
    BrandEquity =~ AWR + QUL + LOY
# structural model
    Credibility ~ H1*Attachment
    SAT ~ H2*Attachment
    BrandEquity ~ H3*Attachment + H4*Credibility + H5*SAT
# allow correlation between mediators
   Credibility ~~ SAT
# Specific indirect effects
  ind_Attachment_via_Credibility := H1 * H4
  ind_Attachment_via_SAT := H2 * H5
# Total indirect effect
  total_ind_Attachment := (H1 * H4) + (H2 * H5)
# Total effect
  total_Attachment := H3 + total_ind_Attachment
"
# run the model
sem_result <- lavaan::sem(
  model,
  data = all_items,
  estimator = "ML",
  se = "bootstrap",
  bootstrap = 2000
)

# extract standardised results
```

Estimated path		95% CI
Direct effects		
Attachment → Credibility	0.71	[0.63, 0.8]
Attachment → SAT	0.53	[0.44, 0.62]
Attachment → BrandEquity	0.26	[0.11, 0.41]
Credibility → BrandEquity	0.23	[-0.07, 0.53]
SAT → BrandEquity	0.60	[0.39, 0.82]
Indirect and total effects		
ind_Attachment_via_Credibility	0.16	[-0.05, 0.38]
ind_Attachment_via_SAT	0.32	[0.19, 0.44]
total_ind_Attachment	0.48	[0.35, 0.62]
total_Attachment	0.74	[0.67, 0.82]

Table 8: Parameter estimates of the hypothesised model - From synthetic data.

```r
std_results <- lavaan::standardizedSolution(sem_result)
# Keep only structural and indirect effects
std_structural <- std_results[
  std_results$op %in% c("~", ":="),
]
# Keep only regressions (~) and defined parameters (:=)
std_results <- std_results[std_results$op %in% c("~", ":="), ]
# Get bootstrap CI from parameterEstimates
boot_pe <- lavaan::parameterEstimates(
  sem_result,
  standardized = TRUE,
  ci = TRUE,
  boot.ci.type = "bca.simple"
)
boot_pe <- boot_pe[boot_pe$op == ":=", ]
```

In this paper, we confine ourselves to the main theoretical model shown in *Panel A*. Parameter estimates of the hypothesised model are summarised in Table 8. We can see that not all the figures are the exactly the same as in the original, but they are very close and the conclusions are the same.

A graphic representation of the structural component of the SEM model created by the synthetic data is presented in Figure 2.

6 Conclusion

This exercise set out to see if we can use functions in the *LikertMakeR* package to "reproduce" raw data based on summary statistics presented in a published report. We find that results are surprisingly strong - producing data that give similar output to those in the original report, and giving the same conclusions. Of course we cannot expect the data to give exactly the same results - after all, they are artificial and subject to rounding of the reported output.

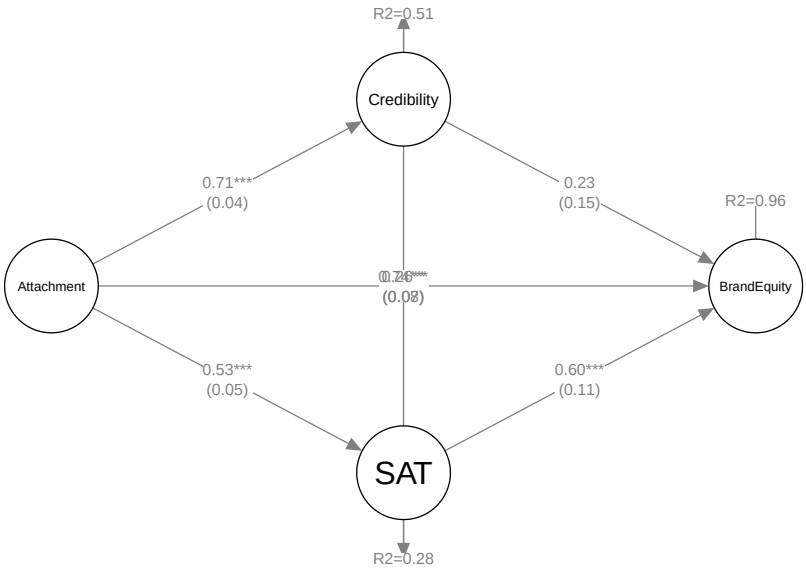

Figure 2: Structural model of Synthetic data - reproducing original data.

6.1 What we have learned

This reconstruction demonstrates that synthetic Likert-scale data can be generated from published summary statistics with high fidelity.

Even though the original raw data are unavailable, the reconstructed dataset reproduces:

- the reported scale distributions
- the correlation structure between constructs
- the reliability estimates
- the results of the structural equation model

This makes synthetic reconstruction useful for:

- teaching statistical methods
- replicating published analyses
- conducting methodological simulations when raw data are unavailable.

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